

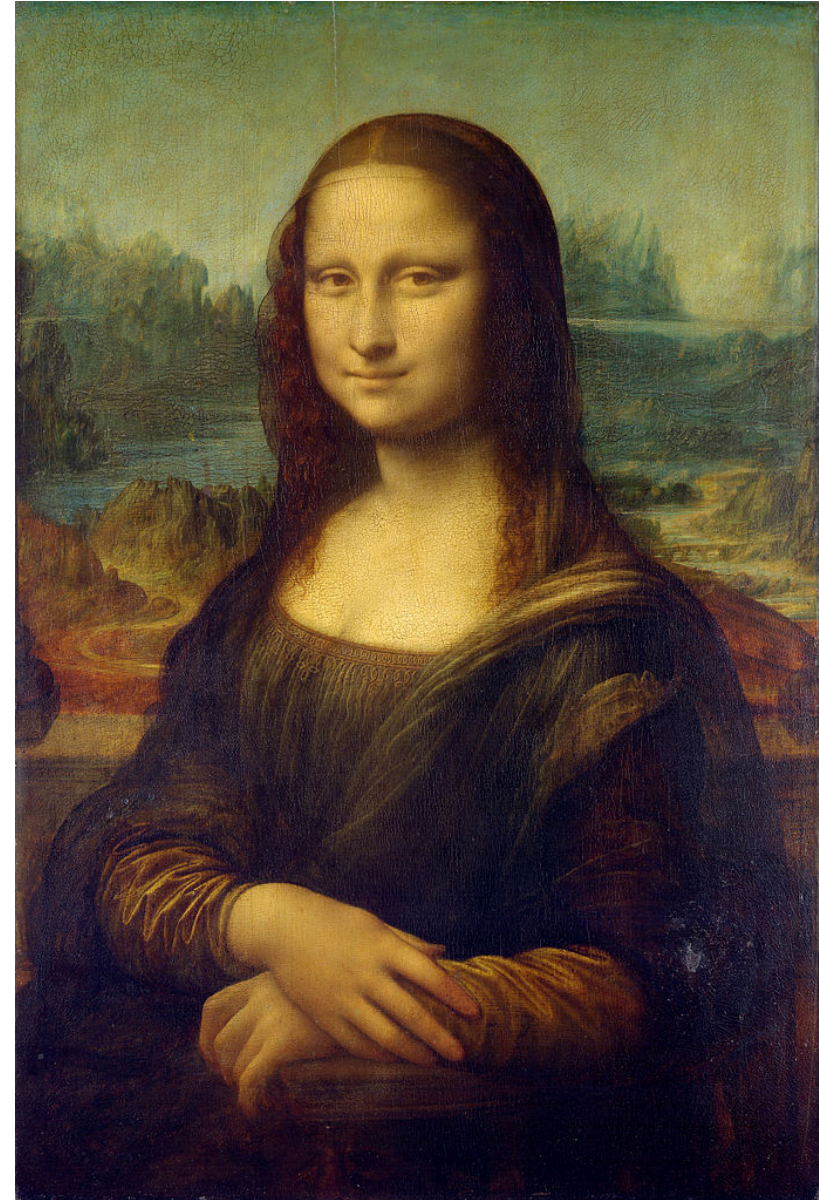
Geographic Data Science

Vector

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(Geo)visualisation

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“Data graphics visually display measured quantities by means of the combined use of points, lines, a coordinate system, numbers, symbols, words, shading, and color.”

The Visual Display of Quantitative Information. Edward R. Tufte.

Visualization

By encoding information visually, they allow to present large amounts of numbers in a meaningful way. If well made, visualizations provide leads into the processes underlying the graphic.

The Visual Display of Quantitative Information. Edward R. Tufte.

Geovisualization

Tufte (1983)

“The most extensive data maps [...] place millions of bits of information on a single page before our eyes. No other method for the display of statistical information is so powerful”

MacEachren (1994)

“Geographic visualization can be defined as the use of concrete visual representations –whether on paper or through computer displays or other media—to make spatial contexts and problems visible, so as to engage the most powerful human information processing abilities, those associated with vision.”

Geovisualization

- Not to replace the human in the loop, but to augment her/him.
- Augmentation through engaging the pattern recognition capabilities that our brain inherently has.
- Combines cartography, infovis and statistics

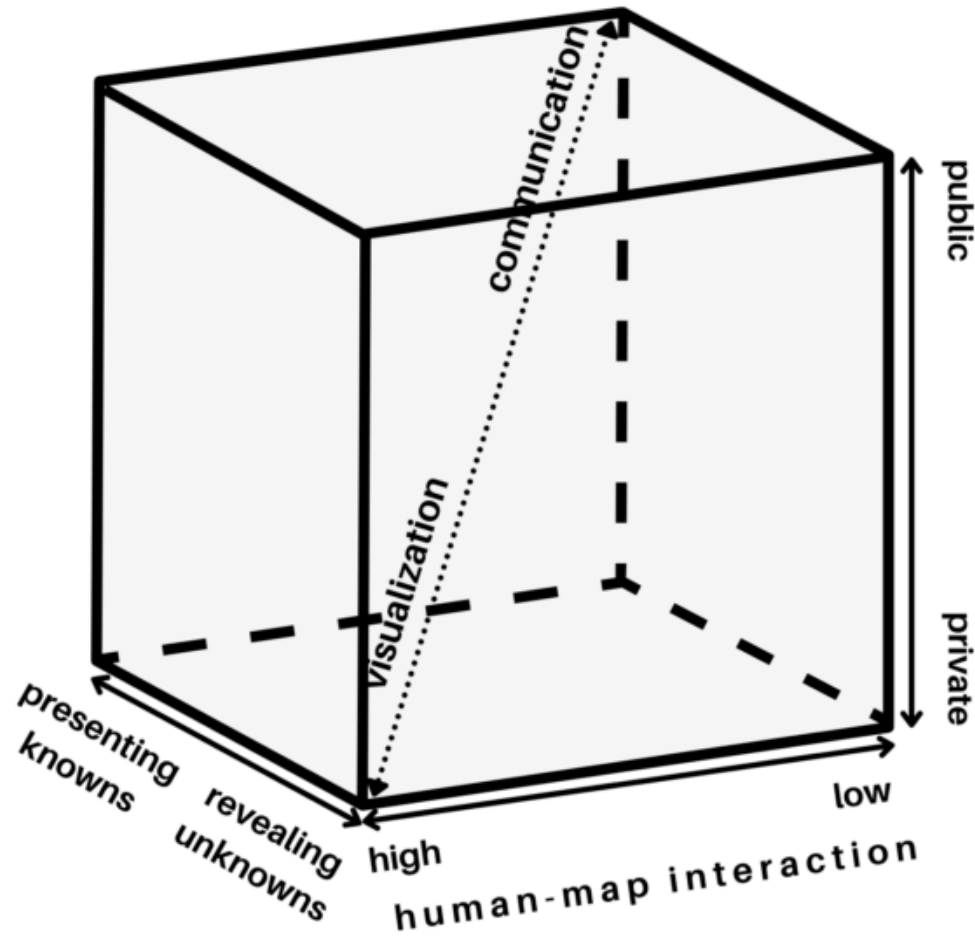
A map for everyone

Maps can fulfill several needs, looking very different depending on the end-goal.

MacEachren & Kraak (1997) identify three main dimensions:

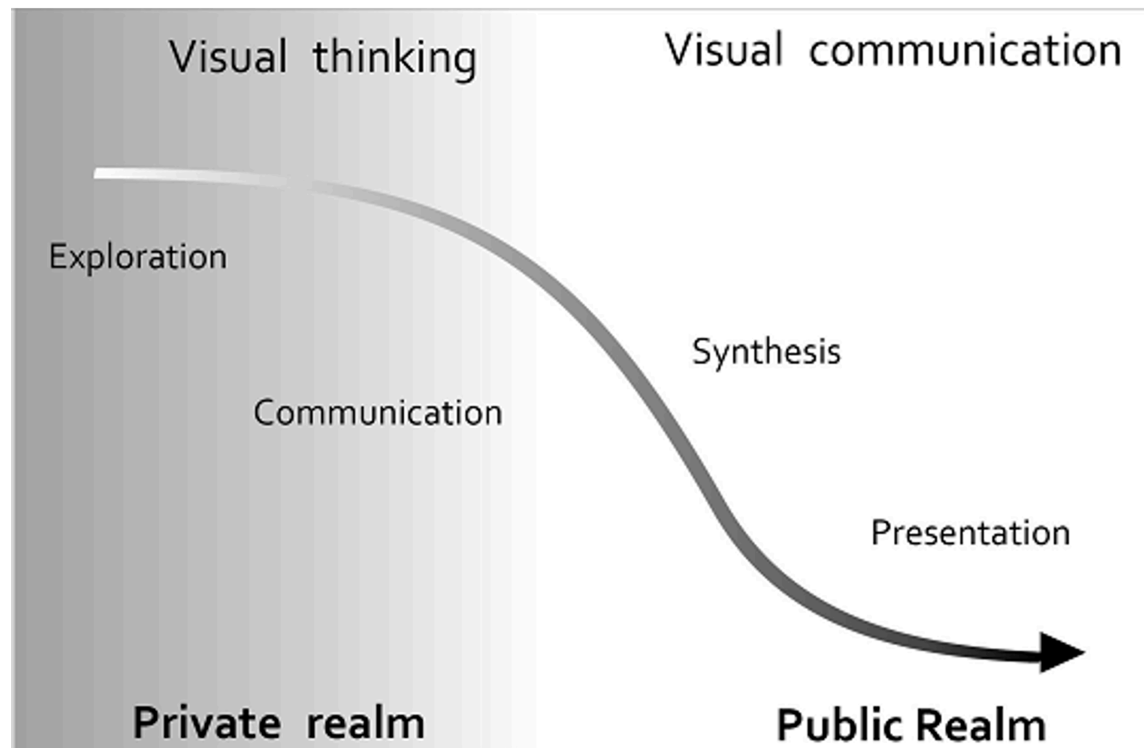
- Knowledge of what is being plotted
- Target audience
- Degree of interactivity

MacEachren & Kraak (1997)



DiBiase's (1990) “Swoopy”

Translating numbers into a (visual) language that the human brain “speaks better”



Exploratory Visualization

“forces us to notice what we never expected to see” (Tukey 1977: vi)

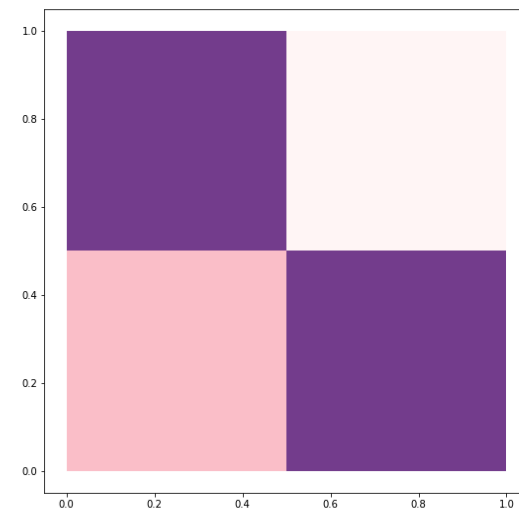
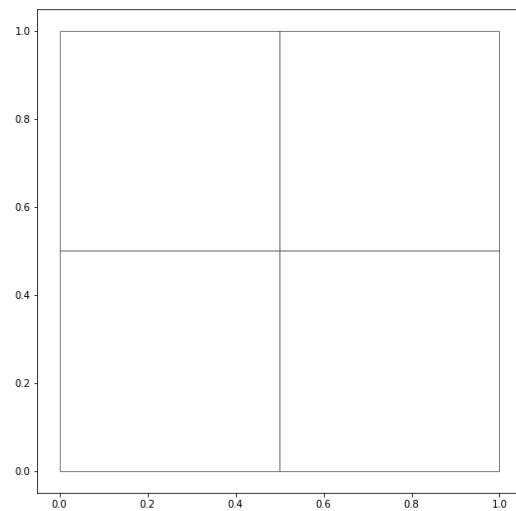
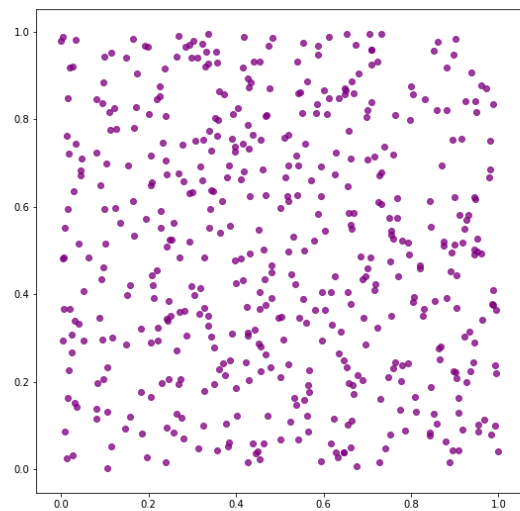
- Mostly for ourselves in the course of the research process.
- Many, quick and dirty, and rather unattractive graphs.

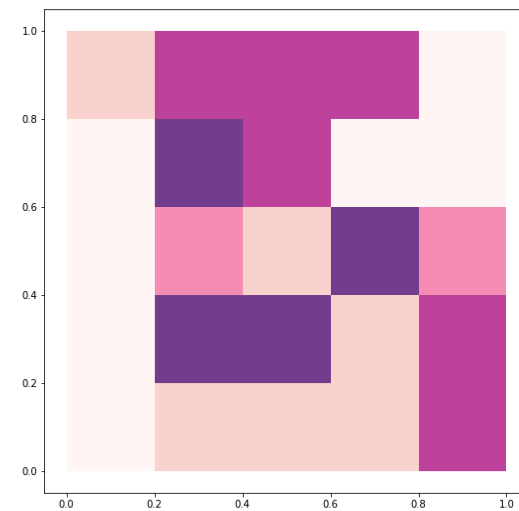
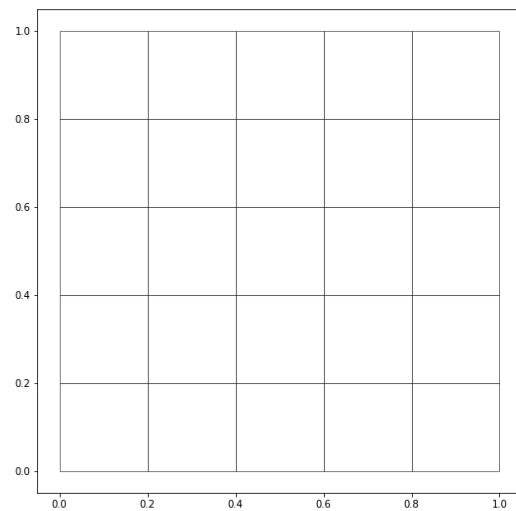
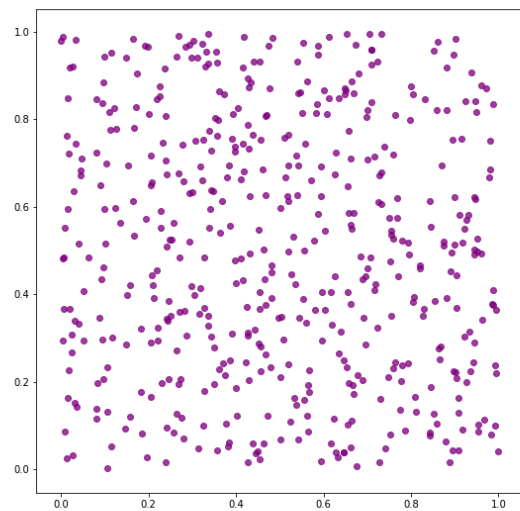
Explanatory Visualization

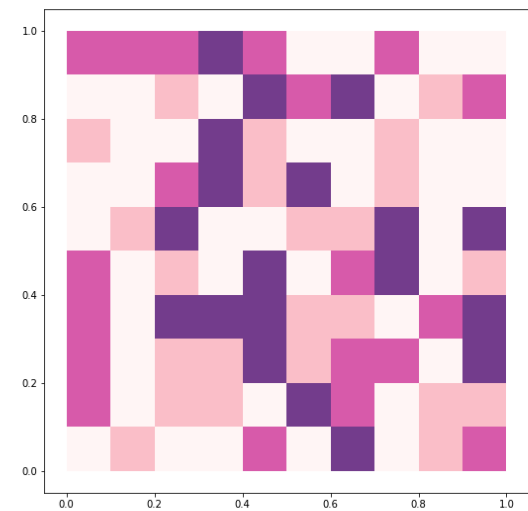
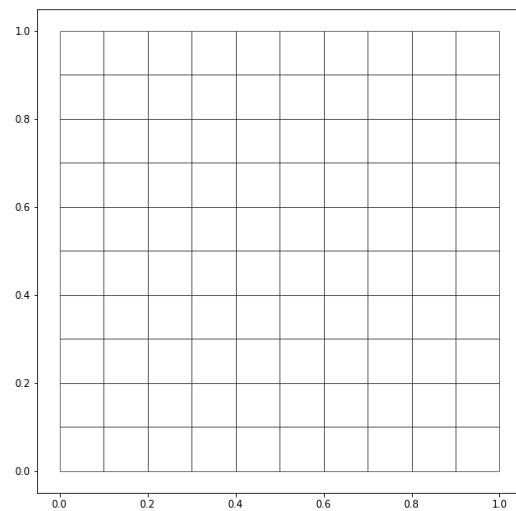
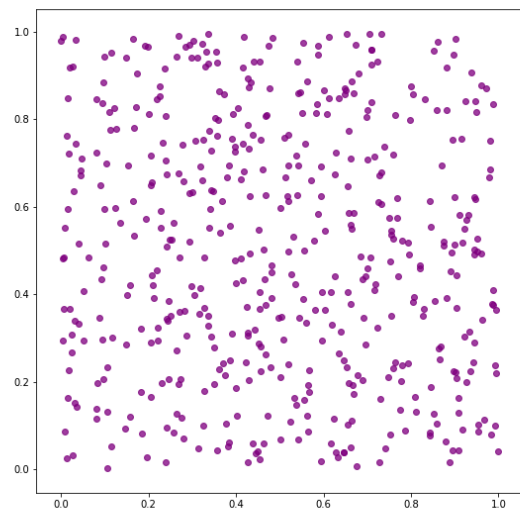
“forces readers to see the information the designer wanted to convey” (Kosslyn 1994: 271)

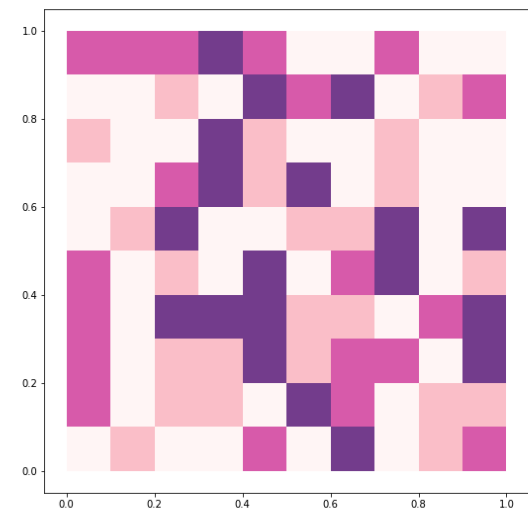
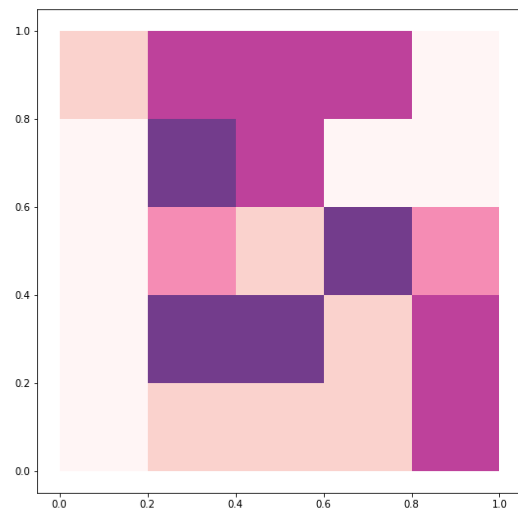
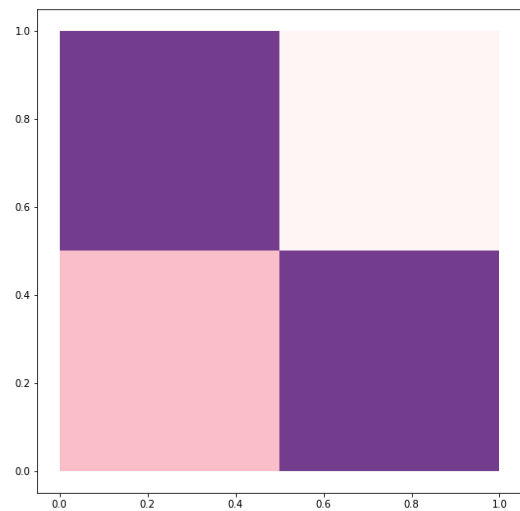
- Mostly for others after the research is completed.
- Few, carefully crafted, and attractive graphs.

Modifiable Area Unit Problem (Openshaw, 1984)









MAUP

Scale and delineation mismatch between:

- Underlying process (e.g. individuals, firms, shops)
- Unit of measurement (e.g. neighborhoods, regions, etc.)
- In some cases, it can seriously mislead analysis on aggregated data (e.g. [Flint](#))

Always keep MAUP in mind when exploring aggregated data!!!

Choropleths

Choropleths

Thematic map in which values of a variable are encoded using a color gradient of some sort

- Counterpart of the histogram

Both allows us to gage the distribution of a variable

- **Values** are classified into specific **colours**: value → bin
- **Information loss** as a trade off for simplicity

Key decision to be made why a given value is a specific colour!

Classification choices

- N. of bins
- How to bin?
- Colours

How many bins?

- Trade-off: detail vs cognitive load
- Exact number depends on purpose of the map
- Usually not more than 12

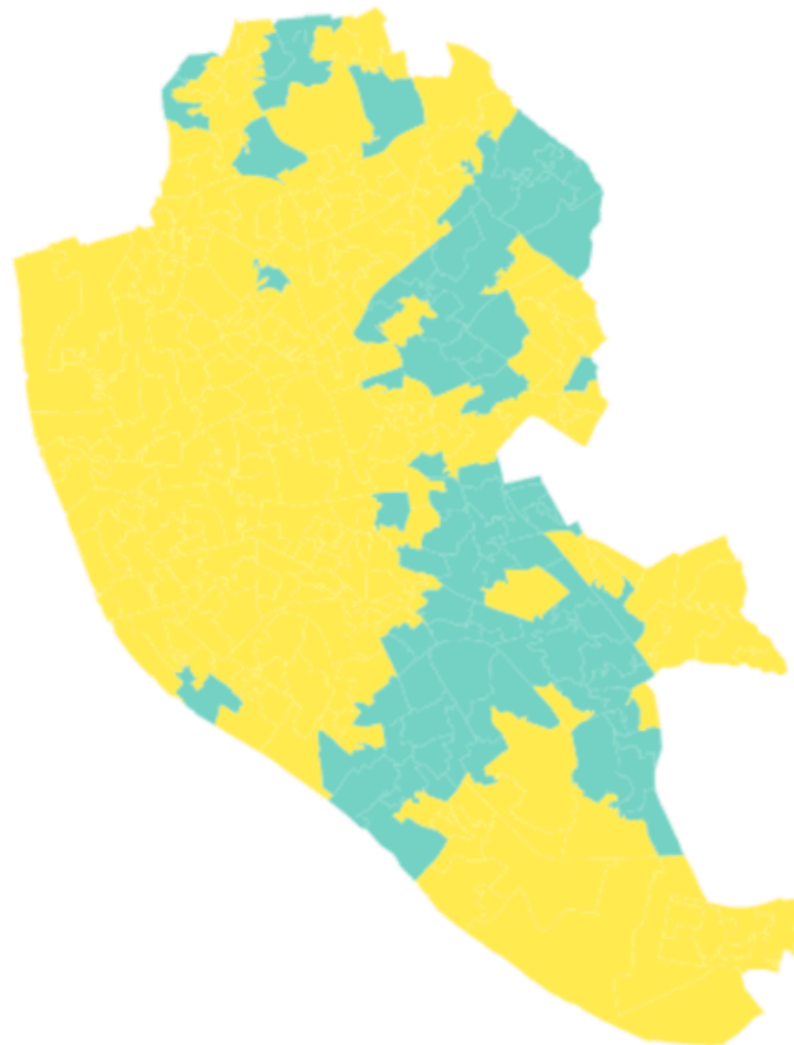
How do we bin?

Essentially a statistical problem

Unique values

- Categorical data
- No gradient (reflect it with the colour scheme!!!)
- Examples: Religion, country of origin...

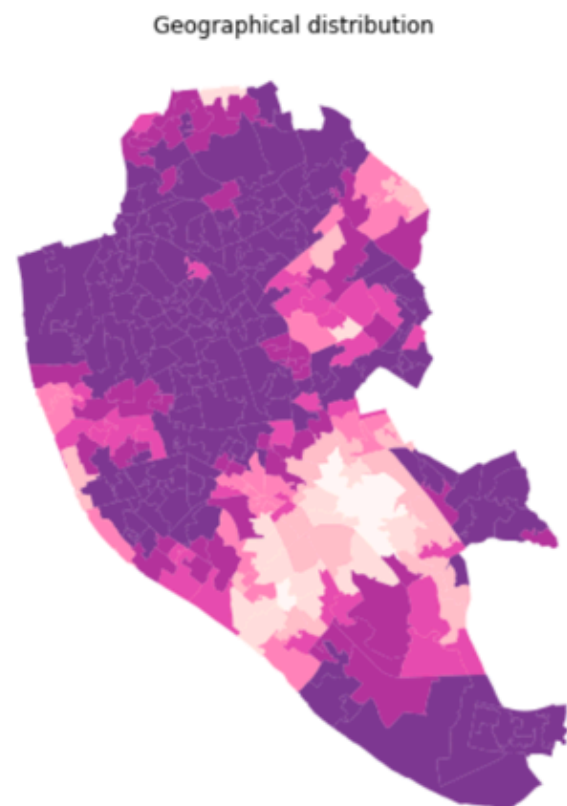
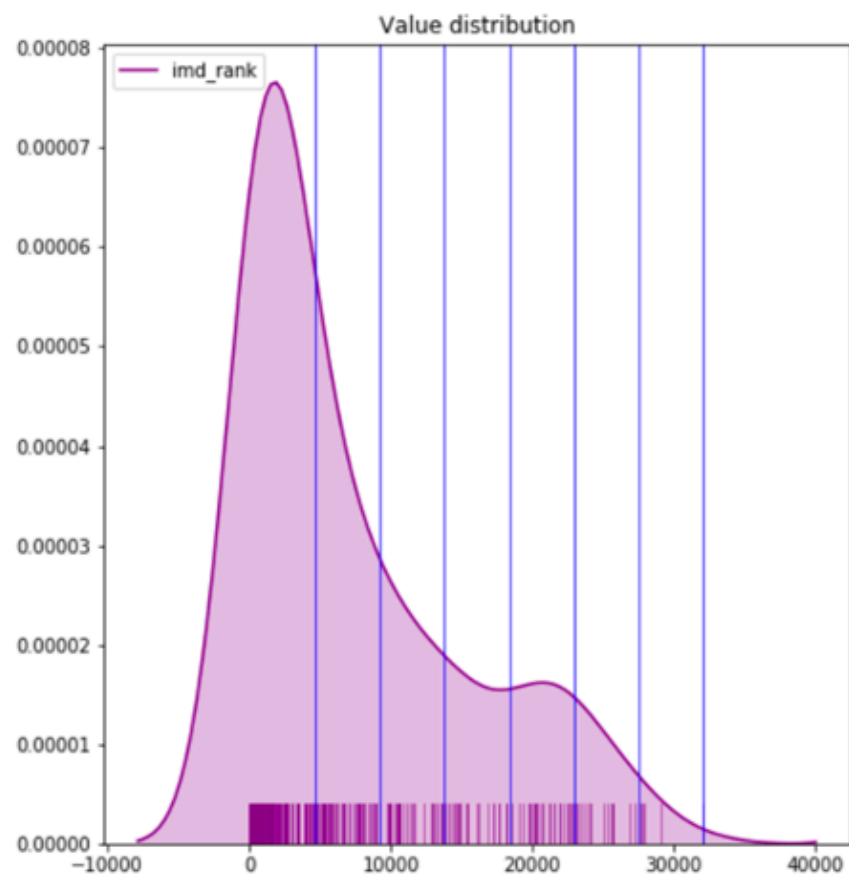
Status Majority



Equal interval (continuous)

- Take the value span of the data to represent and split it equally
- Splitting happens based on the numerical value
- Gives more weight to outliers if the distribution is skewed

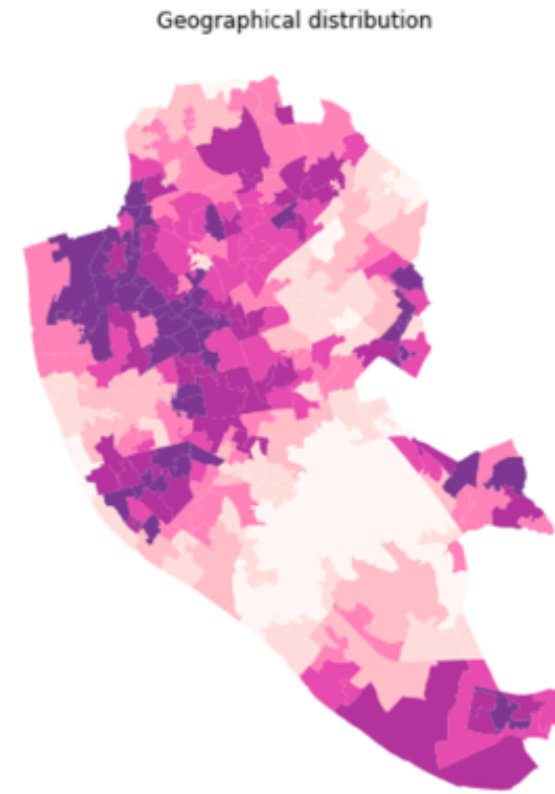
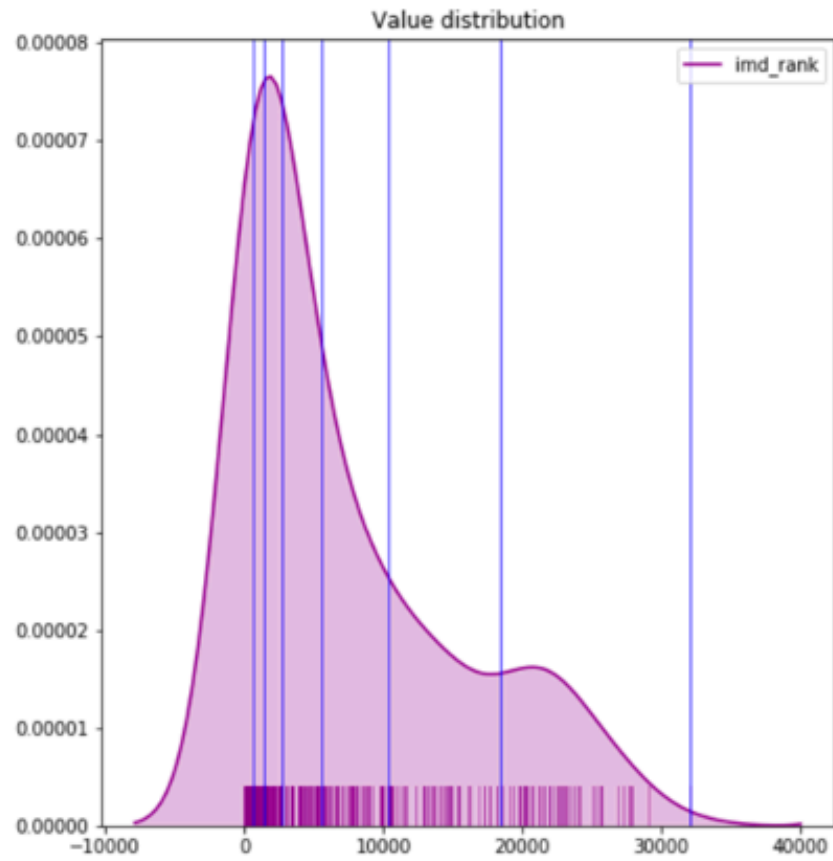
equal_interval



Quantile

- Regardless of numerical values, split the distribution keeping the same amount of values in each bin
- Splitting based on the rank of the value
- If distribution is skewed, it can put very different values in the same bin

quantiles



Different type of algorithms will optimize for different types of splits

- Fisher-Jenks
- Natural breaks
- Outlier maps: box maps, std. maps...

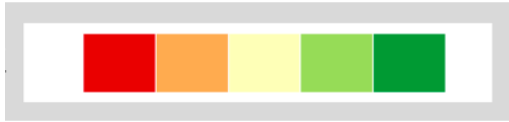
Some involve some fairly fancy statistics.

Colour palette

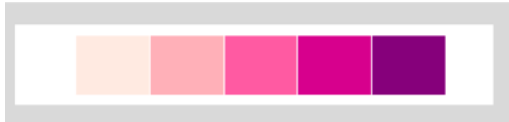
Categories, non-ordered



Graduated, sequential



Graduated, divergent



TIP: check ColorBrewer for guidance

Tips

- Think of the purpose of the map
- Explore by trying different classification alternatives
- Combine (geo)visualisation with other statistical devices

Beyond the classic choropleth

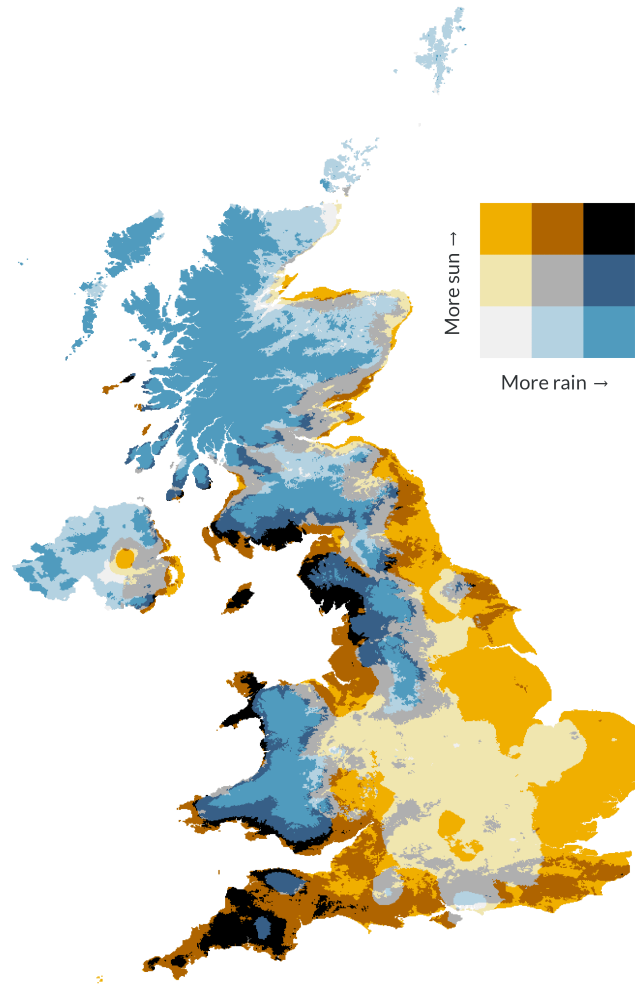
Bivariate choropleths

Show **two variables at once**, using a 3×3 (or similar) grid of colours instead of a single-colour gradient

- Each axis of the grid = one variable's bins
- The colour a region ends up with encodes *both* — e.g. dark = high on both, a single hue = high on one but not the other
- Trade-off: powerful, but the legend takes real effort for a reader to decode — use sparingly, for a genuinely joint story

The ☀️ and the ☁️

Annual hours of sunshine vs. total precipitation in 2021

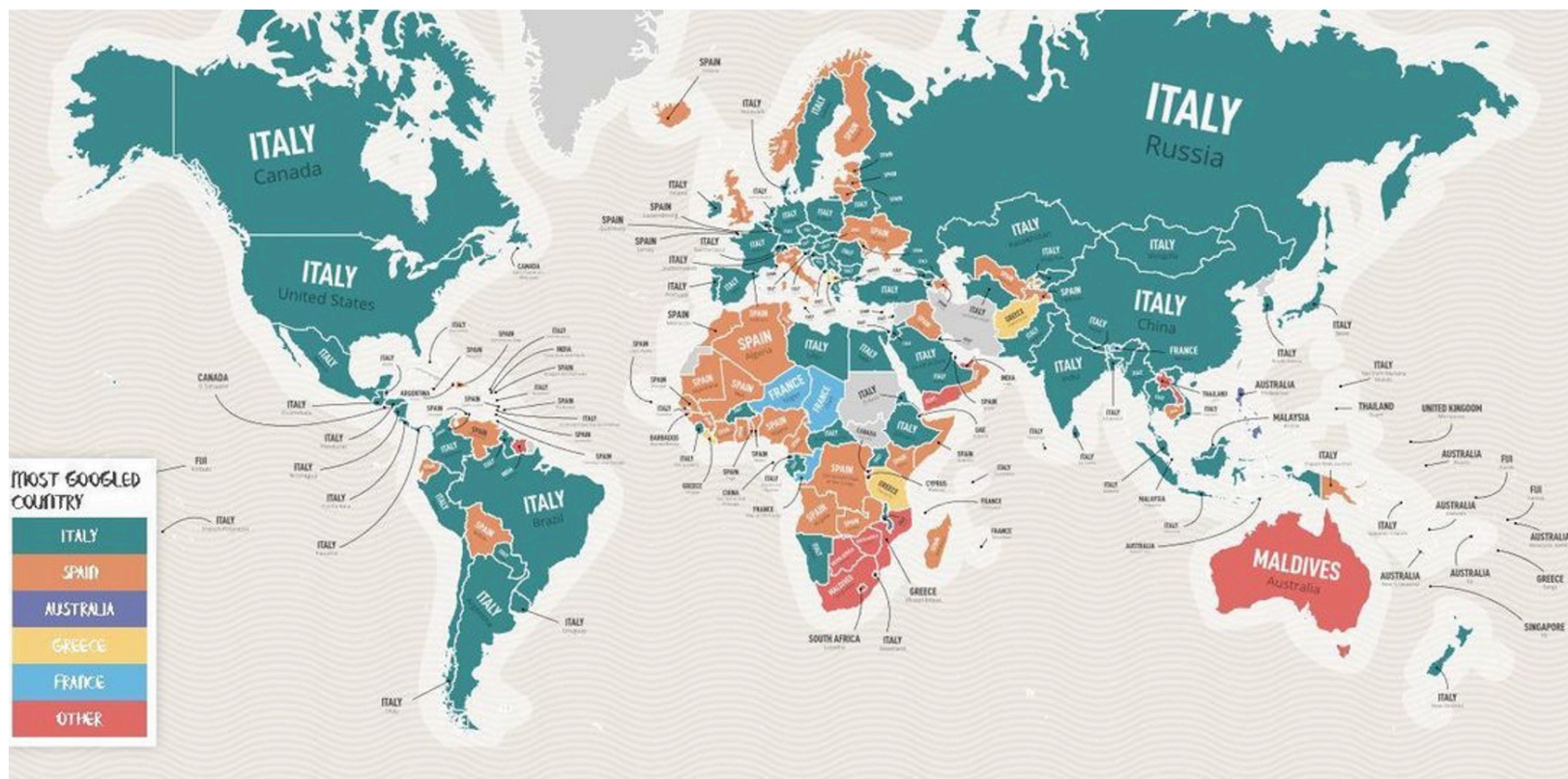


Data from Met Office/Hollis et al./CEDA
Plot by @VictimOfMaths

For more inspiration

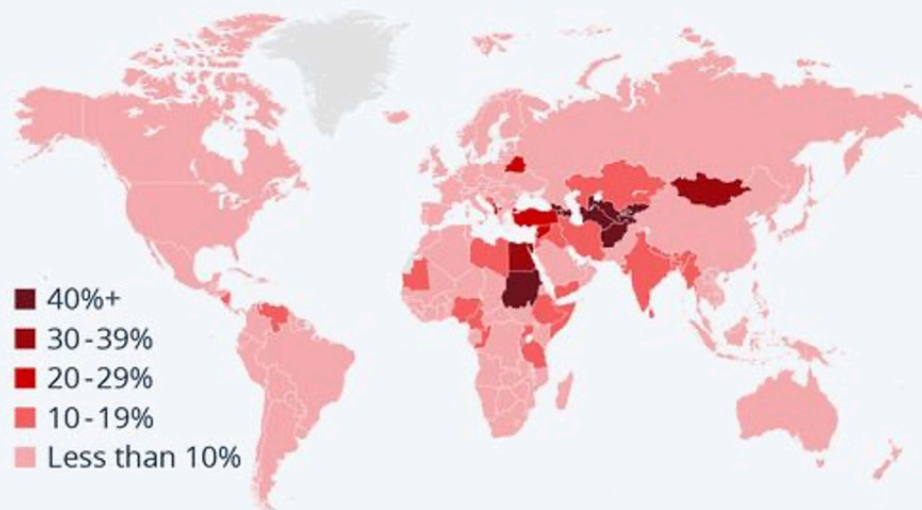
Browse the **#30DayMapChallenge** galleries — a different map type/technique set as a daily prompt each November, thousands of real entries across every style imaginable (points, lines, choropleths, 3D, GeoAI, and more):

30daymapchallenge.com/galleries



Where Food Imports Are Affected by the Ukraine Crisis

Share of food imports per country affected by export restrictions elsewhere (in percent of calories)



As of August 1, 2022. Excludes exports held back from Ukraine.

Source: International Food Policy Research Institute



statista

GLOBAL *Beer* CONSUMPTION

By Country

People have been brewing beer since ancient Mesopotamia nearly 7,000 years ago. Today, it's still one of the most highly consumed beverages across the globe. But beer consumption varies greatly, depending on the country.

Where is beer drank the most?

In the U.S., Budweiser is the most popular beer. In a recent beer ranking survey by TOP agency, Budweiser ranked #1 in 23 of the 50 states.

The Czech Republic is the top country by beer consumption per capita.



China is the top country for beer consumption. In 2020, people in China guzzled more than 36 million kiloliters of beer, accounting for a whopping 20.3% of global consumption that year.



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Source: Kinn Holdings

Questions



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